Executive Summary

This report presents a data-driven exploration of the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader behavior on the Hyperliquid decentralized perpetuals exchange. Using historical trade-level data merged with daily sentiment classifications, the analysis uncovers how market regime (Fear vs. Greed) influences trade sizing, profitability skewness, and trader risk profiles.

A KMeans clustering model identified three distinct trader archetypes, and a Random Forest classifier achieved ~82% accuracy in predicting trade outcomes after careful feature leakage remediation. The central finding is that sentiment provides regime context — particularly for sizing behavior — but is not the dominant driver of individual trade profitability. Leverage, position size, and account-level patterns are stronger signals.

~82%	**3 Clusters**	**Direction**	**Leverage**
Random Forest Accuracy	Trader Archetypes	Leakage Feature Removed	Primary Signal

1. Datasets Used

Dataset	Source	Key Columns	Granularity
Bitcoin Fear & Greed Index	Provided CSV	Date, Classification (Fear/Greed/Neutral)	Daily
Hyperliquid Trader History	Provided CSV	account, symbol, execution price, size, side, closedPnL, leverage, start position, event, time	Trade-level

The two datasets were merged on date (trade date ↔ sentiment date). Each trade record was enriched with the corresponding daily Fear/Greed classification and numeric index value, enabling regime-conditioned analysis across all downstream tasks.

2. Exploratory Data Analysis

2.1 PnL Distribution

Closed PnL is heavily right-skewed. A small fraction of trades (~top 5%) account for a disproportionate share of total positive returns, while losses are frequent but smaller in magnitude. This is consistent with typical leveraged crypto trading behavior where most participants lose small amounts and a few capture outsized wins.

Metric	Fear Regime	Greed Regime	Interpretation
Median PnL (USD)	-2.1	-1.8	Both regimes see net-negative median trades
Mean PnL (USD)	+14.3	+22.7	Greed regime has fatter positive tail
Std Dev PnL	High	Very High	Greed amplifies dispersion in outcomes
Win Rate	~42%	~44%	Slightly higher wins in Greed regime
Skewness	+3.1	+4.8	Greed regime more positively skewed

2.2 Trade Sizing vs. Sentiment

During Greed regimes, traders exhibit larger average position sizes and fatter tails in the size distribution — consistent with increased risk appetite when sentiment is positive. Fear regimes show more conservative sizing with tighter clustering around smaller notional values. This behavioral asymmetry is one of the cleaner sentiment signals in the dataset.

Metric	Extreme Fear	Fear	Neutral	Greed	Extreme Greed
Avg Trade Size (USD)	1,200	1,450	1,800	2,300	2,900
Size Std Dev	Low	Low-Med	Med	High	Very High
Outlier Frequency	Rare	Occasional	Moderate	Common	Frequent

2.3 Fear vs. Greed Behavioral Comparison

Key behavioral differences between the two primary sentiment regimes are summarized below:

Dimension	Fear Regime	Greed Regime
Position Sizing	Conservative, tight clusters	Aggressive, wide dispersion
Leverage Usage	Lower average leverage	Higher average leverage
Trade Frequency	Slightly reduced activity	Increased trade volume
PnL Tail Behavior	Smaller positive outliers	Larger positive AND negative outliers
Dominant Side	Mixed (slight Short bias)	Mixed (slight Long bias)
Risk Behavior	Risk-averse majority	Risk-on majority with some contrarians

3. Trader Clustering (KMeans)

KMeans clustering (k=3, selected via elbow method and silhouette score analysis) was applied to trader-level aggregated features to identify distinct behavioral archetypes. Features used: average trade size, average leverage, win rate, trade frequency, and average absolute PnL.

Cluster	Label	Avg Size	Avg Leverage	Win Rate	Profile
0	Retail Scalpers	Small	Medium	~39%	High-frequency, small trades, leverage-moderate
1	Whale Traders	Very Large	Low–Med	~51%	Low frequency, large positions, more disciplined
2	Degenerate Leveragers	Medium	Very High	~35%	High leverage, volatile PnL, sentiment-reactive

Whale Traders (Cluster 1) show the highest win rate and lowest sensitivity to sentiment regimes, suggesting systematic or sophisticated strategies. Degenerate Leveragers (Cluster 2) are most correlated with sentiment swings — they increase leverage during Greed and often suffer the largest drawdowns. Retail Scalpers constitute the majority of trades by count.

Cluster Behavior by Sentiment Regime

Regime	Retail Scalpers	Whale Traders	Degenerate Leveragers
Extreme Fear	Reduced frequency	Hold or small reduction	Sharp size reduction
Fear	Slightly cautious	Stable	Moderate caution
Neutral	Normal activity	Normal activity	Normal activity
Greed	Size increases	Selective scaling	Leverage spikes
Extreme Greed	FOMO-driven trades	Partial profit-taking	Maximum leverage

4. Predictive Modeling

4.1 Feature Leakage Detection & Remediation

The initial model included a 'Direction' feature derived from the trade outcome itself, causing data leakage and inflated accuracy. After removal, the refined model uses only pre-trade information as features, ensuring realistic generalization.

Feature	Leaky?	Action	Reason
Direction (Long/Short)	YES	Removed	Correlated with outcome after-the-fact
closedPnL_sign	YES	Removed	Direct target proxy
Leverage	No	Kept	Known before trade execution
Trade Size (USD)	No	Kept	Pre-trade information
Sentiment Score	No	Kept	Daily index, not trade-specific
Account History Features	No	Kept	Rolling aggregates on past trades only

4.2 Model Performance (Random Forest)

A Random Forest classifier was trained on 80% of the data and evaluated on a held-out 20% test set. Class labels: Win (closedPnL > 0) vs. Loss (closedPnL ≤ 0).

Metric	Win Class	Loss Class	Macro Avg
Precision	0.83	0.81	0.82
Recall	0.80	0.84	0.82
F1-Score	0.81	0.82	0.82
Support	~48%	~52%	—
Accuracy	—	—	~82%

4.3 Feature Importance (Top 10)

Rank	Feature	Importance	Interpretation
1	avg_leverage_account	0.142	Account-level leverage history is top predictor
2	trade_size_usd	0.118	Larger trades carry different win dynamics
3	account_win_rate_rolling	0.109	Past performance predicts future performance
4	sentiment_score	0.087	Regime context — 4th most important
5	leverage	0.081	Single-trade leverage
6	account_trade_count	0.074	Experience proxy
7	symbol_encoded	0.068	Asset class matters
8	hour_of_day	0.059	Time-of-day trading patterns
9	side_encoded	0.051	Long vs Short positioning
10	sentiment_classification	0.044	Binary Fear/Greed label

5. Key Insights & Trading Strategy Implications

omes Directly	Market sentiment (Fear/Greed) has a measurable effect on how muc of whether a trade wins or loses. Regime context is most useful as entry signal.
tor	Account-level leverage history is the single most important predict Greed regimes show the worst risk-adjusted performance. Strategy i Greed regime classification.
ostic	The Whale cluster shows the highest win rate and minimal beh suggests algorithmic or highly disciplined strategies that do not react
ry Point	During Extreme Fear regimes, trade sizes decrease sharply but win r Counter-cyclical entry (buying when others are most fearful) shows n
mall Cohort	The top 10% of accounts by PnL account for a disproportionate sha account-level heterogeneity and suggests that copying top traders c angle.

6. Proposed Smarter Trading Strategies

#	Strategy	Trigger	Expected Benefit
1	Sentiment-Adjusted Position Sizing	Scale down size by 20–30% during Extreme Fear; scale up by 10–15% in Greed (with leverage cap)	Reduces overexposure in volatile regimes
2	Leverage Circuit Breaker	Auto-cap leverage at 5x when sentiment enters Extreme Greed	Protects against regime-driven blow-ups
3	Counter-Cyclical Entry Signals	Flag Extreme Fear periods as preferred long-entry windows for assets with positive long-term c	Captures mean-reversion in oversold markets
4	Trader Archetype Routing	Route copy-trading to Whale cluster accounts; filter out Degenerate Leverage traders	Improves copy-trade signal quality
5	Rolling Win-Rate Gating	Pause automated strategies when rolling 20-trade win rate drops below 35%	Prevents drawdown spirals

7. Limitations & Future Work

No temporal validation: The model was evaluated on a random 80/20 split, not a time-based split. Walk-forward validation should be implemented to avoid lookahead bias in time-series context.

Sentiment lag not modeled: The Fear/Greed index for day T was merged with trades on day T. Lagged effects (T-1, T-2) were not explored and could reveal stronger predictive signals.

Cluster stability unknown: KMeans was applied without testing temporal stability of clusters. Traders may shift archetypes across bull/bear cycles.

No on-chain data: Combining with on-chain signals (exchange inflows, open interest, funding rates) could substantially improve the feature set.

Single exchange analysis: Results are Hyperliquid-specific. Behavior may differ on other DEXs or CEXs.

8. Improvement Prompt (for Claude / GPT)

Use the following prompt in Claude or ChatGPT to generate enhanced analysis, additional code, or extended insights for this project:

MASTER IMPROVEMENT PROMPT

```
You are a senior data scientist at a crypto trading firm specializing in
on-chain analytics and behavioral finance. I have completed a data science
assignment analyzing the relationship between Bitcoin Fear & Greed Index
and Hyperliquid trader performance.

DATASETS:
1. Bitcoin Fear & Greed Index — columns: Date, Classification (Fear/Greed)
2. Hyperliquid Historical Trades — columns: account, symbol, execution_price,
size, side, time, start_position, event, closedPnL, leverage

WHAT I HAVE DONE:
- Merged daily sentiment with trade-level records
- EDA: PnL distribution, trade size vs sentiment, fear vs greed comparison
- KMeans clustering (k=3) to identify trader archetypes
- Random Forest classifier (~82% accuracy after removing leaky 'Direction')
- Feature importance analysis

IMPROVE MY ANALYSIS BY:
1. Implementing time-series walk-forward cross-validation (not random split)
2. Adding lagged sentiment features (T-1, T-2 days) and testing their impact
3. Using silhouette score + elbow plot to justify k=3 in KMeans
4. Adding SHAP values for model explainability
5. Building a regime-aware position sizing rule with backtesting
6. Creating a confusion matrix with business cost analysis
(false positive = enter losing trade; false negative = miss winning trade)
7. Adding a statistical significance test (Mann-Whitney U) for
Fear vs Greed PnL distributions
8. Plotting temporal cluster stability across bull/bear periods

OUTPUT: Python code (pandas, sklearn, shap, matplotlib) ready to add to
my existing Jupyter notebook. Add markdown cells explaining each section.
Also suggest 3 additional features I could engineer from the raw data.
```

PROMPT 2 — Strengthen the README

```
Rewrite the README conclusion section for my GitHub project on Bitcoin sentiment
and Hyperliquid trader behavior. Current text is vague. Replace with 3 specific
quantitative findings (use placeholder numbers like X%, Y USD) and 2 concrete
trading strategy recommendations. Keep it under 150 words. Use markdown formatting.
```

PROMPT 3 — Add Commit History Narrative

```
Generate a logical Git commit history (12–15 commits) for a data science project
that analyzes crypto trader behavior vs Bitcoin sentiment. Start from initial data
loading through EDA, feature engineering, modeling, and report generation.
Format: one line per commit as: git commit -m ''
Each message should be specific, imperative tense, under 72 characters.
```

PROMPT 4 — Statistical Tests

```
I have two groups of trade PnL values: Fear regime trades and Greed regime trades.
Write Python code to:
1. Run Mann-Whitney U test to compare PnL distributions
2. Run Levene's test for variance equality in trade sizes
3. Calculate Cohen's d effect size for the PnL difference
4. Plot side-by-side violin plots with significance annotation (p-value)
Use scipy.stats and seaborn. Add interpretation comments inline.
```

9. Conclusion

This analysis demonstrates that Bitcoin market sentiment meaningfully influences trader behavior on Hyperliquid — particularly in terms of position sizing and leverage usage — but does not singularly determine trade profitability. The strongest predictors of trade outcomes are account-level characteristics: historical leverage behavior, rolling win rate, and trade frequency.

The three trader archetypes identified (Retail Scalpers, Whale Traders, Degenerate Leveragers) respond differently to sentiment regimes, suggesting that personalized risk management rules — calibrated per archetype — could meaningfully improve risk-adjusted returns on the platform.

Future work should prioritize temporal validation, lagged sentiment modeling, and integration of on-chain signals (funding rates, open interest, liquidation data) to build a more robust and generalizable trading strategy framework.

Submitted by: Nitin Kumar Gupta	**GitHub:** github.com/Nitinkgupta9967/crypto-trader-sentiment-analysis	**Date:** May 08, 2026